INTERPRETABILITY FAILURE MODES on the VCS

MECHANISM BRANCH	METHOD EXHIBITING IT (phase)	F	faithfulness F vs intervention oracle (0 / 0.5 / 1.0)
1 Gradient pathologies — *vanishing on the index · saturation · model-invariance*	Vanilla saliency	0.36	
	SmoothGrad	0.36	
	Guided backprop	0.32	
	Grad×Input / DeepLIFT	0.47	
	Integrated Gradients	0.36	
	Expected Gradients	0.19	
2 Sampling / approximation error — *surrogate variance · baseline sensitivity*	LIME	0.64	
	KernelSHAP	0.65	
	RISE	0.54	
3 Correlational confounds — *present ≠ used · spurious tuning · spurious edges*	Tuning curves (A3)	0.21	
	Spike-word / pairwise (A4)	0.25	
	Granger causality (A6)	0.16	
	Linear probing + control (C)	0.21	
	Sparse autoencoder (C)	0.15	
	NMF/PCA dictionaries (C)	0.13	
	Dim-reduction NMF (A7)	0.52	
4 Decomposition non-identifiability — *polysemantic / rotated atoms ≠ true variables*	Single-unit lesions (A2)	0.96	
	Attribution patching (C)	0.48	
	ACDC circuit disc. (C)	0.65	
	Path patching (C)	0.60	
5 Compositional blind-spots — *single-site misses distributed / interacting structure*	*Grad-CAM / Grad-CAM++*	N/A	*does not apply*
	Attention rollout	N/A	*does not apply*
	VIPER	N/A	*does not apply*
0 Architecture-specific N/A (control — not a failure of method) — *needs conv maps / attention / a learned policy — does not apply*			

(1) gradient pathologies
(2) sampling / approximation
(3) correlational confound
(4) decomposition non-id.
(5) compositional blind-spot
(0) N/A — does not apply (control)
low F (pale chip)
high F (dark chip)
95% bootstrap CI (whisker)